2. Digital Logic and Microprocessor

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Syllabus

2.1 Digital logic: Number Systems, Logic Levels, Logic Gates, Boolean algebra, Sum-of-Products Method, Product-of-Sums Method, Truth Table to Karnaugh Map. (AExE0201)

2.2 Combinational and arithmetic circuits: Multiplexetures, Demultiplexetures, Decoder, Encoder, Binary Addition, Binary Subtraction, operation on Unsigned and Signed Binary Numbers. (AExE0202)

2.3 Sequential logic circuit: RS Flip-Flops, Gated Flip-Flops, Edge Triggered Flip-Flops, Mater- Slave Flip-Flops. Types of Registers, Applications of Shift Registers, Asynchronous Counters, Synchronous Counters. (AExE0203)

2.4 Microprocessor: Internal Architecture and Features of microprocessor, Assembly Language Programming. (AExE0204)

2.5 Microprocessor system: Memory Device Classification and Hierarchy, Interfacing I/O and Memory Parallel Interface. Introduction to Programmable Peripheral Interface (PPI), Serial Interface, Synchronous and Asynchronous Transmission, Serial Interface Standards. Introduction to Direct Memory Access (DMA) and DMA Controllers. (AExE0205)

2.6 Interrupt operations: Interrupt, Interrupt Service Routine, and Interrupt Processing. (AExE0206)

2.1 Digital logic: Number Systems, Logic Levels, Logic Gates, Boolean algebra, Sum-of-Products Method, Product-of-Sums Method, Truth Table to Karnaugh Map. (AExE0201)

Number System

- Binary
- Octal
- Decimal
- Hexadecimal

- Conversions

POSITIVE AND NEGATIVE LOGIC

POSITIVE LOGIC:

- When we use binary 1 for high voltage and binary 0 for low voltage then it is called positive logic.

NEGATIVE LOGIC:

- When we use binary 0 for high voltage and binary 1 for low voltage then it is called Negative logic.

Truth Table

A	Y=A’
1	0
0	1

Negative Logic NOT gate

Truth Table

A	B	Y=A+B
1	1	1
1	0	0
0	1	0
0	0	0

Negative Logic OR gate

Truth Table

A	B	Y=A.B
1	1	1
1	0	1
0	1	1
0	0	0

Negative Logic AND gate

- Positive logic AND gate is equivalent to Negative logic OR gate and vice versa.
- Positive logic NAND gate is equivalent to Negative logic NOR gate and vice versa.
- Positive logic XOR gate is equivalent to Negative logic XNOR gate and vice versa.

Logic Gates

- Basic Gates (NOT, AND, OR)
- Universal Gates (NAND, NOR)
- Exclusive Gates (XOR , XNOR)

Boolean Algebra

- Boolean Algebra is used to analyze and simplify the digital (logic) circuits. It uses only the binary numbers i.e. 0 and 1. It is also called as Binary Algebra or logical Algebra. Boolean algebra was invented by George Boole in 1854.
- Boolean Laws

Commutative law

Any binary operation which satisfies the following expression is referred to as commutative operation.

(i) A.B = B. A (ii) A + B = B + A

Boolean Algebra

Associative law

This law states that the order in which the logic operations are performed is irrelevant as their effect is the same.

(i) $(A.B).C = A.(B.C)$ (ii) $(A + B) + C = A + (B + C)$

Distributive law

Distributive law states the following condition.

$$A.(B + C) = A.B + A.C$$

AND law

These laws use the AND operation. Therefore they are called as **AND** laws.

(i) $A.0 = 0$ (ii) $A.1 = A$

(iii) $A.A = A$ (iv) $A.\overline{A} = 0$

Boolean Algebra

OR law

These laws use the OR operation. Therefore they are called as **OR** laws.

(i) $A + 0 = A$ (ii) $A + 1 = 1$

(iii) $A + A = A$ (iv) $A + \overline{A} = 1$

INVERSION law

This law uses the NOT operation. The inversion law states that double inversion of a variable results in the original variable itself.

$$\overline{\overline{A}} = A$$

DE MORGANS THEOREM

Theorem 1

$$\overline{A.B} = \overline{A} + \overline{B}$$

NAND = Bubbled OR

Theorem 2

$$\overline{A + B} = \overline{A}.\overline{B}$$

NOR = Bubbled AND

FOR MCQ: https://www.sanfoundry.com/discrete-mathematics-questions-answers-de-morgan-laws/

BOOLEAN ALGEBRA

Dual Theorem:

- Starting with a Boolean relation, we can derive another Boolean relation, called its dual by the following steps:
  - Changing each OR sign into AND sign
  - Changing each AND sign into OR sign
  - Complementing all 0's and 1's

Example:

S.N.	Given Expression	Dual of given expression
1	A + AB = A	A. (A + B) = A
2	A + A'B = A + B	A. (A' + B) = A.B
3	A + A' = 1	A.A' = 0
4	(A + B)(A + C) = A + BC	A.B + A.C = A. (B + C)

STANDARD FORM AND CANONICAL FORM

CANONICAL FORM:

- Max Term
- Min Term

Max Term:

- Each Max term is obtained from an OR logic of n variables with each variable being unprimed if the corresponding bit is zero (0) and primed if one (1).

A	B	C	Max term	designation
0	0	0	A+B+C	M_0
0	0	1	A+B+C’	M_1
0	1	0	A+B’+C	M_2
0	1	1	A+B’+C’	M_3
1	0	0	A’+B+C	M_4
1	0	1	A’+B+C’	M_5
1	1	0	A’+B’+C	M_6
1	1	1	A’+B’+C’	M_7

STANDARD FORM AND CANONICAL FORM

Min Term:

- Each Min term is obtained from an AND logic of n variables with each variable being unprimed if the corresponding bit is one (1) and primed if zero (0).

A	B	C	Min term	designation
0	0	0	A'B'C'	m_0
0	0	1	A'B'C	m_1
0	1	0	A'BC'	m_2
0	1	1	A'BC	m_3
1	0	0	AB'C'	m_4
1	0	1	AB'C	m_5
1	1	0	ABC'	m_6
1	1	1	ABC	m_7

STANDARD FORM AND CANONICAL FORM

STANDARD FORMS:

- In standard form the terms that form the function may contain one, two or any number of literals/ variables. There are two types of standard forms.
  - Sum of Product (SOP)
  - Product of Sum (POS)

Sum of Product (SOP)

- SOP is a Boolean expression containing terms with AND logic of 1 or more literals.
- E.g. F=XYZ + X’YZ + X’Y’Z

Product of Sum(POS)

- POS is a Boolean expression containing terms with OR logic of 1 or more literals.
- E.g. F=(X + Y + Z) (X’ + Y + Z) (X’ +Y’ +Z)

KARNAUGH MAP (K-MAP) (SOP)

- K-MAP is regarded as a diagrammatic or pictorial form of a truth table.
- The map is a diagram made up of squares.
- Each square represents one min/ max term.
- The MAP represents a visual diagram of all possible ways of function, may ne expressed in a standard form.

Basic K-MAP

B \ A		B’	B
		0	1
A’	0	A’B’	A’B
A	1	AB’	AB

Fig: Two variable K-MAP

KARNAUGH MAP (K-MAP) (SOP)

Three variable K-MAP

- There are 8 min terms for 3 binary variables.
- A MAP consists of 8 squares.
- The min terms are arranged not in a binary sequence but in sequence similar to gray code.
- The characteristics of the sequence is that only one bit is changes from one sequence to another.

BC / A		B'C'	B'C	BC	BC'
		00	01	11	10
A'	0	m_0	m_1	m_3	m_2
A	1	m_4	m_5	m_7	m_6

Fig: Three variable K-MAP

KARNAUGH MAP (K-MAP) (SOP)

Four variable K-MAP

- There are 16 min terms for 4 binary variables.
- A MAP consists of 16 squares.

CD / AB		C'D'	C'D	CD	CD'
		00	01	11	10
A'B'	00	m_0	m_1	m_3	m_2
A'B	01	m_4	m_5	m_7	m_6
AB	11	m_{12}	m_{13}	m_{15}	m_{14}
AB'	10	m_8	m_9	m_{11}	m_{10}

Fig: Four variable K-MAP

Simplification:

- One square box represents one min term giving a term of four literals.
- Two adjacent square box represents a term of three literals
- Four adjacent square box represents a term of two literals.
- Eight square box represents one min term giving a term of one literals.
- Sixteen adjacent square box represents a function 1
- Zero square box represents a function 0.

KARNAUGH MAP (K-MAP) (SOP)

DON'T CARE CONDITION:

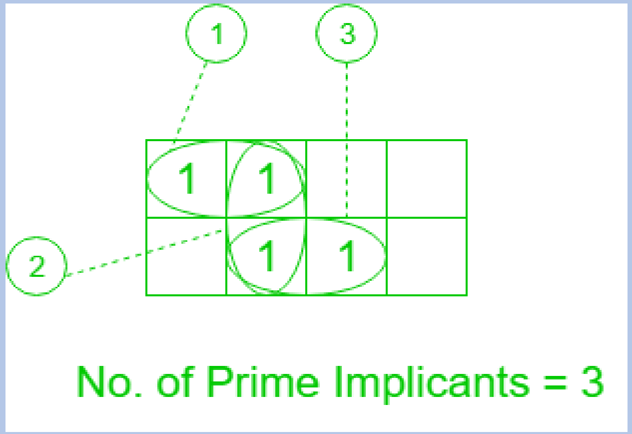
- There are some condition of inputs for which output is not specified and such output does not affect the whole system, which are known as Don't Care condition.
- The Don't care min terms are denoted by 'X' sign.

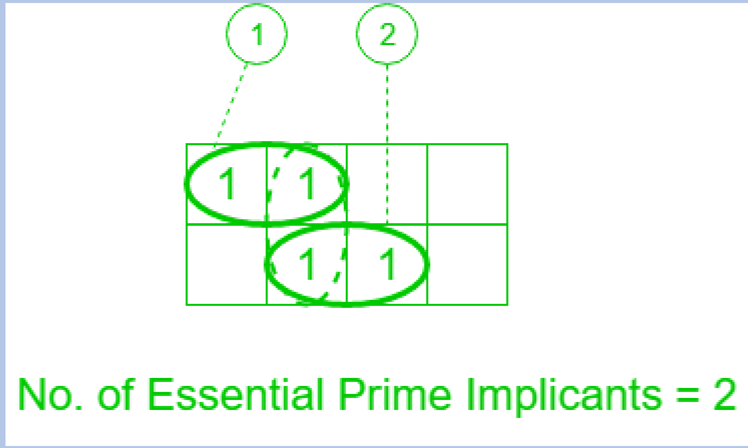
IMPLICANTS IN K-MAP

- **Prime Implicants**
  - A group of square or rectangle made up of bunch of adjacent minterms which is allowed by definition of K-Map are called prime implicants(PI) i.e. all possible groups formed in K-Map.
- **Essential Prime Implicants**
  - These are those sub cubes (groups) which cover at least one minterm that can't be covered by any other prime implicant. Essential prime implicants(EPI) are those prime implicants which always appear in final solution.

KARNAUGH MAP (K-MAP) (SOP)

E.G.





KARNAUGH MAP (K-MAP) (SOP)

K-MAP EXAMPLE:

- Simplify using K-MAP and design a logic circuit.

F (A,B,C,D) = ∑(1,3,7,11,13) and the don't care condition D (A,B,C,D) = ∑(0,2,5)

CD \ AB		C'D'	C'D	CD	CD'
		00	01	11	10
A'B'	00	X	1	1	X
A'B	01	0	X	1	0
AB	11	0	1	0	0
AB'	10	0	0	1	0

A'D

B'CD

BC'D

F = A'D + B'CD + BC'D

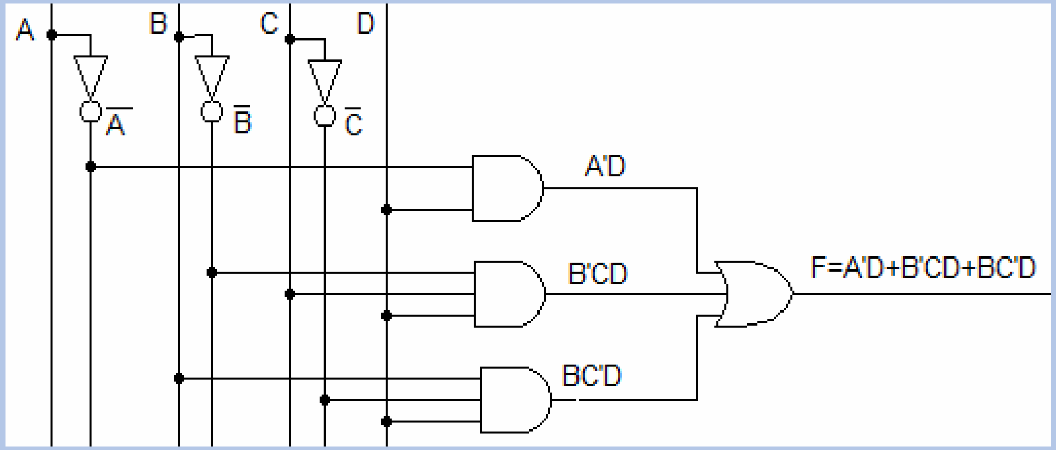


Fig: Combinational Design

Boolean Algebra - MCQ

1. Algebra of logic is termed as ______________
a) Numerical logic
b) Boolean algebra
c) Arithmetic logic
d) Boolean number

2. Boolean algebra can be used ___________
a) For designing of the digital computers
b) In building logic symbols
c) Circuit theory
d) Building algebraic functions

Boolean Algebra - MCQ

3 F(X,Y,Z,M) = X`Y`Z`M`. The degree of the function is ________

a) 2

b) 5

c) 4

d) 1

4 A ________ value is represented by a Boolean expression.

a) Positive

b) Recursive

c) Negative

d) Boolean

Boolean Algebra - MCQ

5 What are the canonical forms of Boolean Expressions?

a) OR and XOR

b) NOR and XNOR

c) MAX and MIN

d) SOM and POM

6 . The __________ of all the variables in direct or complemented from is a maxterm.

a) addition

b) product

c) moduler

d) subtraction

Boolean Algebra - MCQ

7 Which of the following is/are the universal logic gates?

a) OR and NOR

b) AND

c) NAND and NOR

d) NOT

8 The logic gate that provides high output for same inputs __________

a) NOT

b) X-NOR

c) AND

d) XOR

Boolean Algebra – MCQ

9. Which of the following bits is the negation of the bits “010110”?

a) 111001

b) 001001

c) 101001

d) 111111

10. How many bits string of length 4 are possible such that they contain 2 ones and 2 zeroes?

a) 4

b) 2

c) 5

d) 6

Boolean Algebra – MCQ

11 If a bit string contains {0, 1} only, having length 5 has no more than 2 ones in it. Then how many such bit strings are possible?

a) 14

b) 12

c) 15

d) 16

12. If A is "001100" and B is "010101" then what is the value of A (Ex-or) B?

a) 000000

b) 111111

c) 001101

d) 011001

Boolean Algebra – MCQ

13 Which of the following option is suitable, if A is “10110110”, B is”11100000” and C is”10100000”?

a) C=A or B

b) C=~A

c) C=~B

d) C=A and B

14. The Ex-nor of this string “01010101” with “11111111” is?

a) 10101010

b) 00110100

c) 01010101

d) 10101001

Boolean Algebra – MCQ

15

16

Boolean Algebra – MCQ

17 What is the one's complement of this string "01010100"?

a) 10101010

b) 00110101

c) 10101011

d) 10101001

18 What is the 2's complement of this string "01010100"?

a) 10101010

b) 00110100

c) 10101100

d) 10101001

Boolean Algebra – MCQ

19 . If in a bits string of {0,1}, of length 4, such that no two ones are together. Then the total number of such possible strings are?

a) 1

b) 5

c) 7

d) 4

20 . Let A: "010101", B=?, If { A (Ex-or) B } is a resultant string of all ones then which of the following statement regarding B is correct?

a) B is negation of A

b) B is 101010

c) {A (and) B} is a resultant string having all zeroes

d) All of the mentioned

Boolean Algebra – MCQ

21 The terms in SOP are called ___________

a) max terms

b) min terms

c) mid terms

d) sum terms

22 Which of the following is an incorrect SOP expression?

a) x+x.y

b) (x+y)(x+z)

c) x

d) x+y

Boolean Algebra – MCQ

23 A sum of products expression is a product term (min term) or several min terms ANDed together.

a) True

b) False

24. LSI stands for ___________

a) Large Scale Integration

b) Large System Integration

c) Large Symbolic Instruction

d) Large Symbolic Integration

Boolean Algebra – MCQ

25

4. The corresponding min term when x=0, y=0 and z=1.

a) x.y.z'

b) X'+Y'+Z

c) X+Y+Z'

d) x'.y'.z

26

. Which operation is shown in the following expression: (X+Y').(X+Z).(Z'+Y')

a) NOR

b) ExOR

c) SOP

d) POS

Boolean Algebra – MCQ

27 . The number of min terms for an expression comprising of 3 variables?

a) 8

b) 3

c) 0

d) 1

28 The number of cells in a K-map with n-variables.

a) 2n

b) n^2

c) 2^n

d) n

Boolean Algebra – MCQ

29 The output of AND gates in the SOP expression is connected using the _______ gate.

a) XOR

b) NOR

c) AND

d) OR

30. The expression A+BC is the reduced form of ____________

a) AB+BC

b) (A+B)(A+C)

c) (A+C)B

d) (A+B)C

Boolean Algebra – MCQ

31 A Karnaugh map (K-map) is an abstract form of ___________ diagram organized as a matrix of squares.

a) Venn Diagram

b) Cycle Diagram

c) Block diagram

d) Triangular Diagram

32 Each product term of a group, w'.x.y' and w.y, represents the ___________ in that group.

a) Input

b) POS

c) Sum-of-Minterms

d) Sum of Maxterms

Boolean Algebra – MCQ

33 There are ______ cells in a 4-variable K-map.

a) 12

b) 16

c) 18

d) 8

34 The prime implicant which has at least one element that is not present in any other implicant is known as __________

a) Essential Prime Implicant

b) Implicant

c) Complement

d) Prime Complement

Boolean Algebra – MCQ

35 It should be kept in mind that don't care terms should be used along with the terms that are present in ________

a) Minterms

b) Expressions

c) K-Map

d) Latches

36 Product-of-Sums expressions can be implemented using __________

a) 2-level OR-AND logic circuits

b) 2-level NOR logic circuits

c) 2-level XOR logic circuits

d) Both 2-level OR-AND and NOR logic circuits

Boolean Algebra – MCQ

37 Each group of adjacent Minterms (group size in powers of twos) corresponds to a possible product term of the given _________

a) Function

b) Value

c) Set

d) Word

38 Don't care conditions can be used for simplifying Boolean expressions in _________

a) Registers

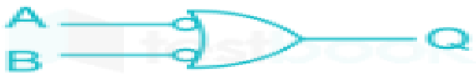
b) Terms

c) K-maps

d) Latches

Boolean Algebra – MCQ

39



From the given arrangement, find Q.

1. $A + B$
2. $A + \overline{B}$
3. $\overline{A} + \overline{B}$
4. More than one of the above

40

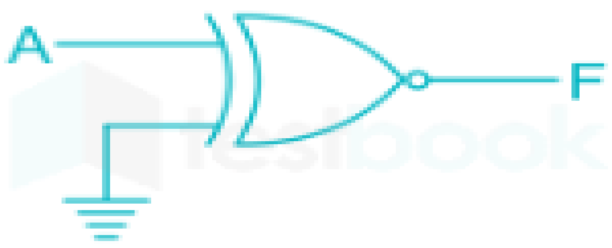
The following truth-table belongs to which one of the four gates-

A	B	X
1	1	0
0	1	0
1	0	0
0	0	1

1. OR
2. NOR
3. XOR
4. More than one of the above

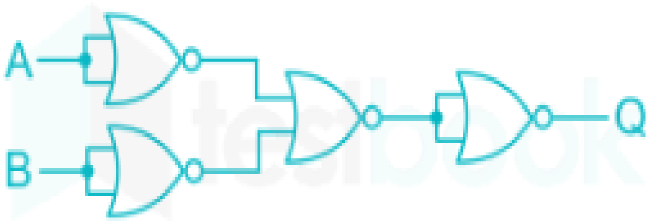
Boolean Algebra – MCQ

41 The output of the logic gate in figure is



1. 0
2. 1
3. $\bar{A}$
4. A

42 The output of logic circuit given below represents ______ gate.

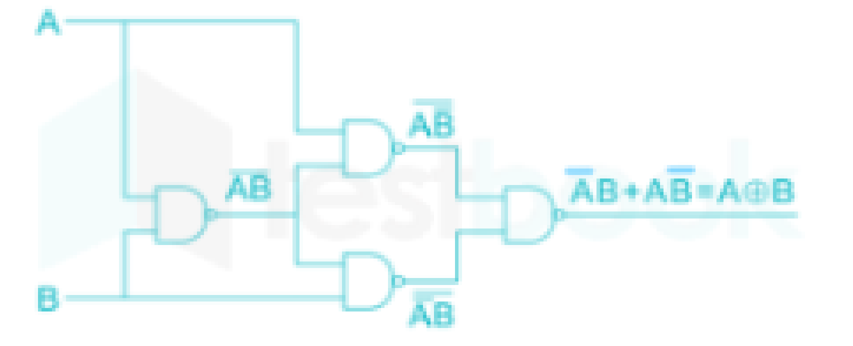


1. OR
2. NOR
3. AND
4. NAND

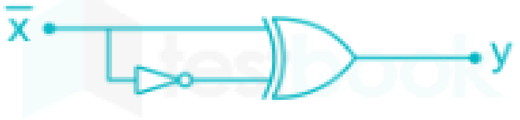
Boolean Algebra – MCQ

43 The minimum number of 2-input NAND gates required to implement a 2-input XOR gate is

1. 4
2. 5
3. 6
4. 7



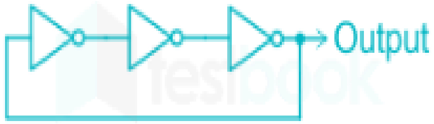
44 The output Y of the logic circuit given below is:-



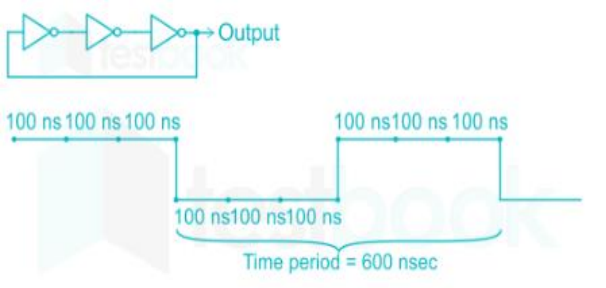
1. 1
2. 0
3. X
4. $\bar{X}$

Boolean Algebra – MCQ

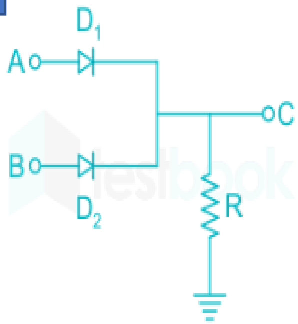
45 What will be the fundamental frequency for the following circuit if each inverter delay is 100 nsec?



1. 1 GHz
2. 0.5 GHz
3. 3.34 MHz
4. 1.67 MHz



46 Which of the following logical operations could be computed by the given network?



1. C = AB
2. C = A + B
3. $C = \overline{AB}$
4. $C = \overline{A + B}$

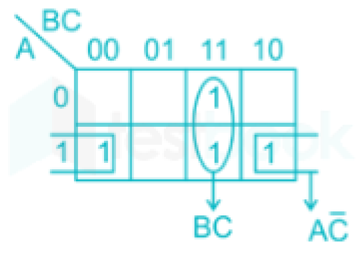
Boolean Algebra – MCQ

47 Boolean expression $AB + A\bar{C} + BC$ simplifies to

1. $BC + A\bar{C}$
2. $AB + A\bar{C} + B$
3. $AB + A\bar{C}$
4. $AB + BC$

48 How many AND and OR gates are required to realise $Y = AB + BC + CD$?

1. 2, 3
2. 2, 1
3. 3, 1
4. 2, 2



$F = BC + A\bar{C}$

Boolean Algebra – MCQ

49 he following hexadecimal number (1E.43)16 is equivalent to

a. $(36.506)_8$

b. $(36.206)_8$

c. $(35.506)_8$

d. $(35.206)_8$

50 How many bits are needed to store one BCD digit?

a. 2 bits

b. 4 bits

c. 3 bits

d. 1 bit

THANK YOU